



CS 540 Introduction to Artificial Intelligence
Linear Models & Linear Regression
University of Wisconsin-Madison

Spring 2025

Announcements

- **Homeworks:**
 - HW4 deadline on **Monday Feb. 24th at 11:59 PM**
- Class roadmap:

ML Linear Regression

Machine Learning: K -
Nearest Neighbors & Naive
Bayes

Machine Learning: Neural
Networks I (Perceptron)

Machine Learning: Neural
Networks II

Supervised Learning

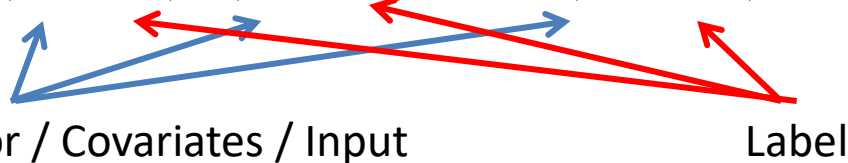
Outline

- Supervised Learning with Linear Models
 - Parameterized model, model classes, linear models, train vs. test
- Linear Regression
 - Least squares, normal equations, residuals, logistic regression

Supervised Learning

Supervised learning:

- Make predictions, classify data, perform regression
- Dataset: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)$



- Goal: find function $f : X \rightarrow Y$ to predict label on **new** data



indoor



outdoor

Regression

- Continuous label $y \in \mathbb{R}$
- Squared loss function $\ell(f(x), y) = (f(x) - y)^2$
- Finding f that minimizes the empirical risk

prediction label



Train loss, we want to minimize this

$$\frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i)$$

Average

The lower, the better

Functions/Models

The function f is usually called a model

- Which possible functions should we consider?

- One option: **all functions**

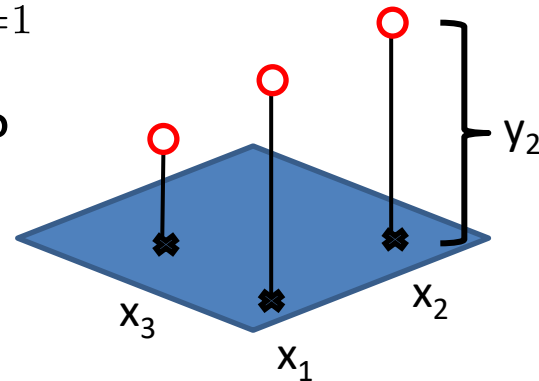
- Not a good choice. Consider

$$f(x) = \sum_{i=1}^n 1\{x = x_i\} y_i$$

- Training loss: **zero**. Can't do better!

- How will it do on x not in the training set?

(cannot generalize)



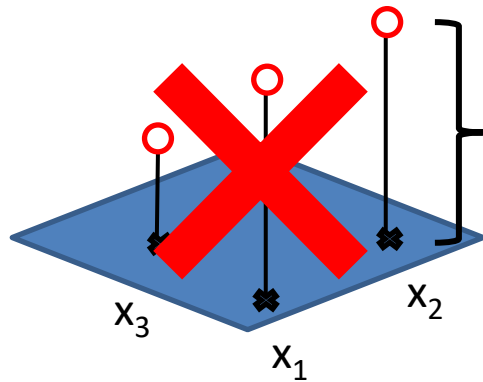
Functions/Models

★ Don't want all functions

- Instead, pick a specific class
- Parametrize it by weights/parameters
- **Example:** linear models

★
$$f(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_d x_d = \theta_0 + x^T \theta$$

Weights/ Parameters



Training The Model

- Parametrize it by weights/parameters

- ✱ • Minimize the loss

Best parameters = best function f  $\min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i)$

$$= \frac{1}{n} \sum_{i=1}^n \ell(\theta_0 + x_i^T \theta, y_i)$$

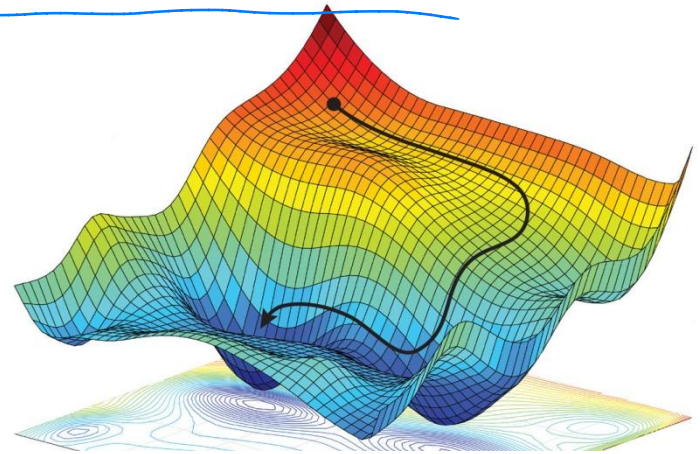
 Linear model class f

$$= \frac{1}{n} \sum_{i=1}^n (\theta_0 + x_i^T \theta - y_i)^2$$

 Square loss

How Do We Minimize?

- Need to solve something that looks like $\min_{\theta} g(\theta)$
- Generic optimization problem; many algorithms
 - A popular choice: **stochastic gradient descent (SGD)**
- Most algorithms iterative:
find some sequence of
points heading towards the
optimum



Train vs Test

Now we've trained, have some f parametrized by θ

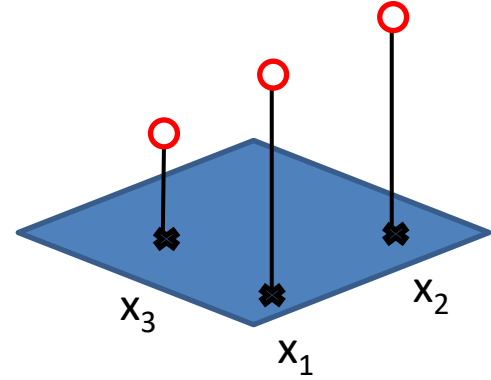
- Train loss is small $\rightarrow f$ predicts most x_i correctly
- How does f do on points not in training set? “Generalizes!”
- To evaluate this, reserve a **test** set. Do **not** train on it!

$$\underbrace{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)}_{\text{Training Data}} \quad \underbrace{(\mathbf{x}_{n+1}, y_{n+1}), \dots, (\mathbf{x}_{n+p}, y_{n+p})}_{\text{Test Data}}$$

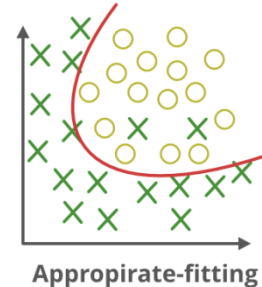
Train vs Test

Use the test set to evaluate f

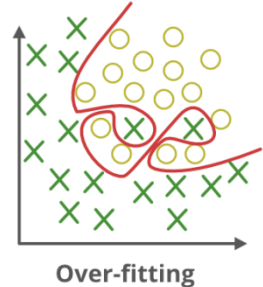
- Why? Back to our “perfect” train function
- Training loss: 0. Every point matched perfectly
- How does it do on test set? **Fails completely!**



- Test set helps detect **overfitting**
 - ✱ Overfitting: too focused on train points
 - “Bigger” class: more prone to overfit
 - Need to consider **model capacity**



GFG



Break & Quiz

Q 1.1: When we train a model, we are

- A. Optimizing the parameters and keeping the features fixed.
- B. Optimizing the features and keeping the parameters fixed.
- C. Optimizing the parameters and the features.
- D. Keeping parameters and features fixed and changing the predictions.

Break & Quiz

Q 1.1: When we train a model, we are

- **A. Optimizing the parameters and keeping the features fixed.**
- B. Optimizing the features and keeping the parameters fixed.
- C. Optimizing the parameters and the features.
- D. Keeping parameters and features fixed and changing the predictions.

Break & Quiz

Q 1.1: When we train a model, we are

- **A. Optimizing the parameters and keeping the features fixed.**
- B. Optimizing the features and keeping the parameters fixed)
(Feature vectors x_i don't change during training).
- C. Optimizing the parameters and the features. (Same as B)
- D. Keeping parameters and features fixed and changing the predictions. (We can't train if we don't change the parameters)

Break & Quiz

- **Q 1.2:** You have trained a classifier, and you find there is significantly **higher** loss on the test set than the training set. What is likely the case?
 - A. You have accidentally trained your classifier on the test set.
 - B. Your classifier is generalizing well.
 - C. Your classifier is generalizing poorly.
 - D. Your classifier is ready for use.

Break & Quiz

- **Q 1.2:** You have trained a classifier, and you find there is significantly **higher** loss on the test set than the training set. What is likely the case?
- A. You have accidentally trained your classifier on the test set.
- B. Your classifier is generalizing well.
- **C. Your classifier is generalizing poorly.**
- D. Your classifier is ready for use.

Break & Quiz

- **Q 1.2:** You have trained a classifier, and you find there is significantly **higher** loss on the test set than the training set. What is likely the case?
- A. You have accidentally trained your classifier on the test set. **(No, this would make test loss lower)**
- B. Your classifier is generalizing well. **(No, test loss is high means poor generalization)**
- **C. Your classifier is generalizing poorly.**
- D. Your classifier is ready for use. **(No, will perform poorly on new data)**

Break & Quiz

- **Q 1.3:** You have trained a classifier, and you find there is significantly **lower** loss on the test set than the training set. What is likely the case?
 - A. You have accidentally trained your classifier on the test set.
 - B. Your classifier is generalizing well.
 - C. Your classifier is generalizing poorly.
 - D. Your classifier needs further training.

Break & Quiz

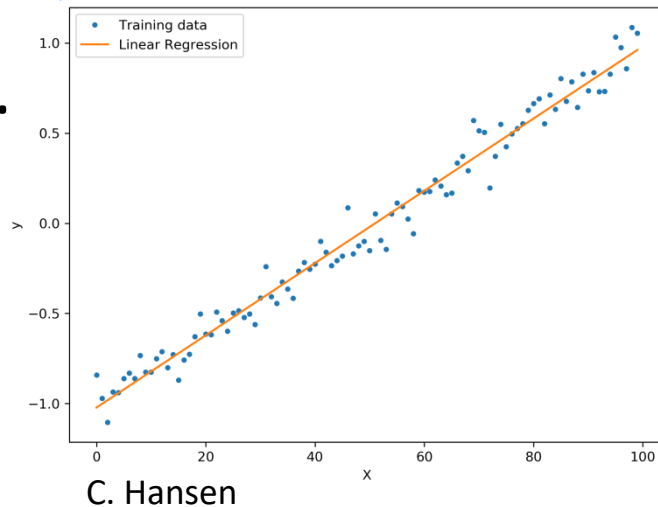
- **Q 1.3:** You have trained a classifier, and you find there is significantly **lower** loss on the test set than the training set. What is likely the case?
- **A. You have accidentally trained your classifier on the test set.** (This is very likely, loss will usually be the lowest on the data set on which a model has been trained)
- B. Your classifier is generalizing well.
- C. Your classifier is generalizing poorly.
- D. Your classifier needs further training.



Linear Regression

Simplest type of regression problem.

- **Inputs:** $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)$
 - x 's are vectors, y 's are scalars.
 - “Linear”: predict a linear combination of x components + intercept



$$\star f(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_d x_d = \theta_0 + x^T \theta$$

- Want: parameters θ

↓
Intercept : $y = kx + b$

Linear Regression Setup

Problem Setup

- Goal: figure out how to minimize square loss
- Let's organize it. Train set $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)$
- ★ – Since $f(x) = \theta_0 + x^T \theta$, use a notational trick by augmenting feature vector with a constant dimension of 1:

$$x = \begin{bmatrix} 1 \\ x \end{bmatrix}$$

- Then, with this one more dimension we can write (θ contains θ_0 now)

$$f(x) = x^T \theta$$

$$x = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} \rightarrow \therefore x^T \theta = \theta_0 + x_1 \theta_1 + \dots + x_n \theta_n$$

(let it be in the form $y = kx$)

Linear Regression Setup

Problem Setup

- Train set $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)$
- Take train features and make it a $n \times (d+1)$ matrix, and y a vector:

Design matrix: X

$$X = \begin{bmatrix} x_1^T \\ \dots \\ x_n^T \end{bmatrix}$$

$$y = \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix}$$

- Then, the empirical risk is $\frac{1}{n} \|X\theta - y\|^2$ ★

↓
Also called the "loss"

Finding The Estimated Parameters

(** We want to find θ that minimizes the loss:*)

Have our loss: $\frac{1}{n} \|X\theta - y\|^2 \rightarrow = (X\theta)^2 - 2X\theta y + y^2$

- Could optimize it with SGD, etc... $\frac{d}{d\theta} = 0 \rightarrow 2X^T X \theta - 2X^T y = 0$
- But the minimum also has a closed-form solution (vector calculus):

Hat: indicates an estimate

$$\hat{\theta} = (X^T X)^{-1} X^T y$$

Not always invertible...

“Normal Equations”

How Good are the Estimated Parameters?

Now we have parameters $\hat{\theta} = (X^T X)^{-1} X^T y$

- How good are they?
- Predictions are $f(x_i) = \hat{\theta}^T x_i = ((X^T X)^{-1} X^T y)^T x_i$
- Errors (“residuals”)

$$|y_i - f(x_i)| = |y_i - \hat{\theta}^T x_i| = |y_i - ((X^T X)^{-1} X^T y)^T x_i|$$

- If data is linear, residuals are 0. Almost never the case!
- **Mean squared error** on a test set

$$\frac{1}{m} \sum_{i=n+1}^{n+m} (\hat{\theta}^T x_i - y_i)^2$$

Linear Regression → Classification?

What if we want the same idea, but y is 0 or 1?

- Need to convert the $\theta^T x$ to a probability in $[0,1]$



★
$$p(y = 1|x) = \frac{1}{1 + \exp(-\theta^T x)}$$

linear

← Logistic function

Why does this work?

- If $\theta^T x$ is really big, $\exp(-\theta^T x)$ is really small $\rightarrow p$ close to 1
- If really negative exp is huge $\rightarrow p$ close to 0

“Logistic Regression”

Break & Quiz

Q 2.1: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

What are the labels, number of points (n), and dimension of the features (d)?

- A. labels are 2 and 4; $n=3$, and $d=2$.
- B. labels are 2 and 4; $n=2$, and $d=3$.
- C. labels are $[-1, 0, 1]$ and $[2, 3, 1]$; $n=2$, and $d=4$.
- D. labels are 2 and 3; $n=4$, and $d=2$.

Break & Quiz

Q 2.1: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

What are the labels, number of points (n), and dimension of the features (d)?

- A. labels are 2 and 4; $n=3$, and $d=2$.
- **B. labels are 2 and 4; $n=2$, and $d=3$.**
- C. labels are $[-1, 0, 1]$ and $[2, 3, 1]$; $n=2$, and $d=4$.
- D. labels are 2 and 3; $n=4$, and $d=2$.

Break & Quiz

Q 2.1: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

What are the labels, number of points (n), and dimension of the features (d)?

- A. labels are 2 and 4; $n=3$, and $d=2$.
- **B. labels are 2 and 4; $n=2$, and $d=3$.**
- C. labels are $[-1, 0, 1]$ and $[2, 3, 1]$; $n=2$, and $d=4$.
- D. labels are 2 and 3; $n=4$, and $d=2$.

There are two data points, each x has 3 features, and the labels are the y -values.

Break & Quiz

Q 2.2: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. Predict $\hat{y}$ for $x = [1, 10, 1]$

- A. 15
- B. 9
- C. 13
- D. 21

Break & Quiz

Q 2.2: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. Predict $\hat{y}$ for $x = [1, 10, 1]$

- A. 15
- B. 9
- **C. 13**
- D. 21

Break & Quiz

Q 2.2: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. Predict $\hat{y}$ for $x = [1, 10, 1]$

- A. 15
- B. 9
- **C. 13**
- D. 21

$$\hat{y} = 1 * \beta_0 + 1 * \beta_1 + 10 * \beta_2 + 1 * \beta_3 = 13$$

Break & Quiz

Q 2.3: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. What is the mean squared error (MSE) on the training set?

- A. 9
- B. $13/2$
- C. $25/2$
- D. 25

Break & Quiz

Q 2.3: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. What is the mean squared error (MSE) on the training set?

- A. 9
- B. $13/2$
- C. **$25/2$**
- D. 25

Break & Quiz

Q 2.3: You have a dataset for regression given by $(x_1, y_1) = ([-1, 0, 1], 2)$ and $(x_2, y_2) = ([2, 3, 1], 4)$.

We have the weights $\beta_0 = 0, \beta_1 = 2, \beta_2 = 1, \beta_3 = 1$. What is the mean squared error (MSE) on the training set?

- A. 9
- B. $13/2$
- C. $25/2$
- D. 25

Compute the predicted label for each data point, then compute the squared error for each data point, then take the mean error of the two points:

$$\hat{y}_1 = -1 * \beta_1 + 0 * \beta_2 + 1 * \beta_3 = -1$$
$$\ell(\hat{y}_1, y_1) = (-1 - 2)^2 = 9$$

$$\hat{y}_2 = 2 * \beta_1 + 3 * \beta_2 + 1 * \beta_3 = 8$$
$$\ell(\hat{y}_2, y_2) = (8 - 4)^2 = 16$$

$$\text{MSE} = (16 + 9) / 2 = 25 / 2$$

Reading

- Linear regression, logistic regression, stochastic gradient descent by Prof. Zhu
<https://pages.cs.wisc.edu/~jerryzhu/cs540/handouts/regression.pdf>